

QUESTION 1: WHAT IS THE APPLICATION ALL ABOUT?

Permit to Work (PTW) System — High-Hazard Authorization Platform

THE CORE SAFETY EQUATION

PTW = Formal Permission + Hazard Identification + Safety Controls + Authorized Sign-Off

 Permit Receiver

(The Job Leader)

Supervisor/Foreman who applies for the permit, manages crew safety, and remains physically present at the site until job completion.

 Permit Issuer

(The Plant Owner)

Operations/Area Engineer who inspects the area, isolates energy/piping (LOTO), verifies safety, and signs authorization.

 PTW Coordinator

(Traffic Controller)

Central master controller who maintains the Permit Register, tracks expiry, and ensures no two dangerous jobs clash (SIMOPS).

 Safety Officer (EHS)

(Safety Guardian)

Conducts gas testing (LEL/toxic), inspects safety gear, endorses permits, and exercises instant Stop Work Authority.

QUESTION 2: WHAT ARE THE FEATURES OF THIS APPLICATION?

Color-Coded Permits, Real-Time Controls & Workforce Tracking

1. Standard Color-Coded Work Permits

 Hot Work (Red)

For sparks, fire, welding, cutting, grinding.

- Must test gas (0.0% LEL), assign fire watch & cover sewers within 75 ft.

 Cold Work (Light Blue)

Base permit for work without sparks or flames.

- Pipe blinding/de-blinding, formwork, painting, cable pulling.

 Confined Space Entry (Green)

Tanks, vessels, manholes, silos, boilers.

- Tests oxygen, toxic gases; mandates blowers & outside standby guard.

 Electrical & LOTO (Yellow/Orange)

Work on energized or isolated electrical circuits.

- Lockout/Tagout padlocks, zero-energy switch checks, circuit grounding.

 Specialized Permits (Purple/Brown/Black)

- Height (> 1.8m harness), Excavation (underground scan), Night Work.

2. Built-in Safety Governance Features

 4-Category Hazard Profiling

Systematic screening of Conditions (height/space), Exposure (noise/heat), Chemicals (COSHH), and Events (explosion/collapse).

 Pre-Task Safety Talks (TBT)

Compulsory safety briefings held at the worksite and signed by all workers before work begins.

 Crew Acknowledgment Sheet

Digital registry capturing every worker's name, employee ID, and trade role for legal accountability.

 Daily Revalidation Engine

Multi-day permits (up to 6 days) strictly require daily re-inspection and Issuer signature each morning.

QUESTION 3: WHY IS THIS APPLICATION REQUIRED?

Catastrophe Prevention, Energy Isolation & Regulatory Compliance

1. Disaster & Explosion Prevention

Refineries hold explosive gases and volatile hydrocarbons. A single stray spark near an unmonitored line causes catastrophic fires. PTW proves zero flammable gas exists before tools touch equipment.

2. Conflict Detection (No Blind Spots)

Multiple contractor crews work on the same plant simultaneously. PTW deconflicts overlapping zones (e.g. stopping hot welding directly over toxic gas venting).

3. Energy Isolation (LOTO Verification)

Ensures equipment is physically de-energized, depressurized, drained, and locked out so machines cannot start by accident while workers are inside.

4. Shared Multi-Party Responsibility

Worksite safety is never placed on a single person. Issuer, Receiver, and Safety Officers must all jointly inspect and sign off before work starts.

5. Regulatory Compliance & Traceability

Maintains an unalterable audit log of gas readings, safety talks, and approvals to satisfy OSHA, ISO 45001, and corporate safety regulations.

QUESTION 4: HOW DOES THE APPLICATION FLOW END-TO-END?

The 8-Stage Lifecycle from Job Planning to Final Site Handback

Step 1

Permit Request

Receiver submits task scope, tools, JSA, and crew list.

Step 2

Technical Review

Issuer verifies equipment isolations, valves, and switches.

Step 3

SIMOPS Check

Coordinator registers permit and checks nearby work zones.

Step 4

Safety Inspection

Safety Officer tests air quality (gas test) & endorses.

Step 5

Activation & TBT

Permit placed in site box; crew briefed in Toolbox Talk.

Step 6

Work & Revalidation

Job executed; daily review and signature for multi-day jobs.

Step 7

Site Handback

Tools packed, isolations removed, site cleaned thoroughly.

Step 8

Final Closeout

Issuer & Receiver sign completion; permit archived.

QUESTION 5: WHAT HAPPENS DURING SIMULTANEOUS INCIDENTS?

SIMOPS Conflict Deconfliction & Stop Work Authority (SWA) Protocols

1. SIMOPS (Simultaneous Operations)

• What is SIMOPS?

When two or more teams work at the same time in the same or nearby area (e.g. welding on scaffolding directly above workers on a pipeline).

• Automatic Conflict Lock

The Coordinator checks the master map. If jobs clash (Vertical elevation hazard or Horizontal proximity), the system locks permit activation.

• Mandatory SIMOPS Checklist

Crews must attach a joint checklist establishing physical fire blankets, protective netting, dedicated watchmen, and shared radio channels.

• Joint Sign-Off Required

Both work crew leaders must agree and co-sign before either team can begin.

2. Emergency / Incident Response Protocol

● Step 1: Stop Work Authority (SWA)

ANY worker can stop work immediately without asking permission if gas leaks, bad weather, chemical spills, or alarms occur.

🚨 Step 2: Notify & Sound Alarm

Permit Receiver alerts the Control Room, Issuer, and Safety team. Crew evacuates to safe muster points if needed.

✖ Step 3: All Active Permits Voided

When an emergency alarm rings, ALL active permits in the area are instantly cancelled and declared invalid. No work can restart informally.

🔍 Step 4: Investigate & Closeout

Incident cause is investigated (Root Cause Analysis). Existing permit is officially stamped 'Work Suspended'.

📄 Step 5: New Permit Required

Work can ONLY resume by issuing a BRAND NEW PERMIT with fresh site inspections and gas tests.